



DataPulse360

**Web-Based 360-Degree Feedback and Assessment
System for Management Education 2026**

Functional Specifications

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Introduction

DataPulse360 aims to design and develop a fully functional, secure, and configurable web-based 360-degree feedback system that supports management and leadership education. We seek to deliver a platform that allows instructors to run complete assessment cycles—from student registration to anonymous rater feedback to individualized reports—while ensuring flexibility for different survey instruments each semester. Our goal is to produce a reliable, scalable tool that can be used by Professor Sirkwoo Jin and potentially adopted by other departments. Users will be able to rate themselves on different leadership styles and skills, and receive peer based feedback on those same styles so they can see what their strengths and weaknesses are.

Executive Summary and System Overview

The website will allow for users to login and complete self evaluation surveys that they've been invited to complete. Admin users will be able to create surveys and invite students to review those surveys, while normal users will only be able to invite other students to evaluate surveys they already have access to. Survey access and account permissions will be handled by our secure account system. After students have completed their survey and have had at least three other students complete the survey from the 3rd person, they will be given access to their personal and classwide data results. This data will be made available in the form of bar graphs sent to users via email.

Use Case Scenarios

Scenario 1: Administrator survey creation

Administrators will sign into the website using their preprovided administrator login credentials. Administrator accounts will be created directly by one of the capstone participants, and cannot be created through the website. Once logged in

administrators will navigate to the surveys tab, and select the “add survey” button. This will prompt them to enter a name for the survey, and continue onto creation. Administrators are given a starting content block that allows for two text field inputs and a numerical input field for the number of answers they wish to allow. Administrators may click the “add question” button to add more questions, and the “remove question” button to remove questions. Upon completing the survey to their satisfaction, administrators will click the “save survey” button. This will assign the survey a unique ID and save the survey to the website’s database. Administrators will be able to access their created surveys from the surveys tab, and the codes for each survey will be displayed here. These codes will be distributed to students at the administrator’s discretion.

Scenario 2:

Users accessing and completing surveys:

Users will create an account or log into the website with an existing one. Upon doing so they will navigate to the surveys tab and select the “use invite code” button. This will prompt them for an invite code given to them by the administrator or a fellow student. This website will communicate with its database and add the survey to the user’s account, and make it appear on the user’s website. Users may now select the button that links to the freshly added survey. While in the survey users will need to select 1 answer for each survey question. Upon finishing the survey properly the student will be able to click the “submit survey” button. The server will save the students' inputs in a spreadsheet document stored on the survey’s computer, and will mark the survey as completed for the student to prevent duplicated survey takers. Students will then be prompted to invite three of their peers to complete the survey by providing their email addresses to the website.

Scenario 3:

User’s completing a survey after being invited by another nonadmin user:

After receiving an email address notifying the user they’ve been invited to rate their peers, users will proceed with the same instructions as above. The primary difference will be in the invite code, which will be appended with a 1 at the beginning in order to load the peer rater version of the questions. These results will

not be saved for the user to view, and will only be saved inside of the survey's spreadsheet documentation.

Screen-by-Screen UI Specification

Home: The central hub screen for the website. Links users to everything inside the website, and displays short information about their to-dos.

View Surveys: Allows users to view their existing surveys. Links to the create survey tab.

- Admins can create surveys and view all surveys stored on the website's server
- Users can view the surveys they've been invited to complete by professors, but can only access surveys they have not yet completed.

My Invites: A secondary tab for invites to appear notifying users that they have an email invitation.

- Will include the username of the person who invited them, and the nature of the survey.

Sign Up: Allows users to create an account.

- Creates and stores user's account information in the website's database. Only one account per email address.
- Usernames must be unique

Sign In: Allows users to sign in using a pre-existing account

- Takes username and password as input. An account must already exist within the website's database in order to be accessed.

Personal Results: Shows users their personal results for all of the surveys they've taken.

- Only allows users to view their personal results for posterities sake. Survey wide information will be distributed by email at the survey creator's discretion.

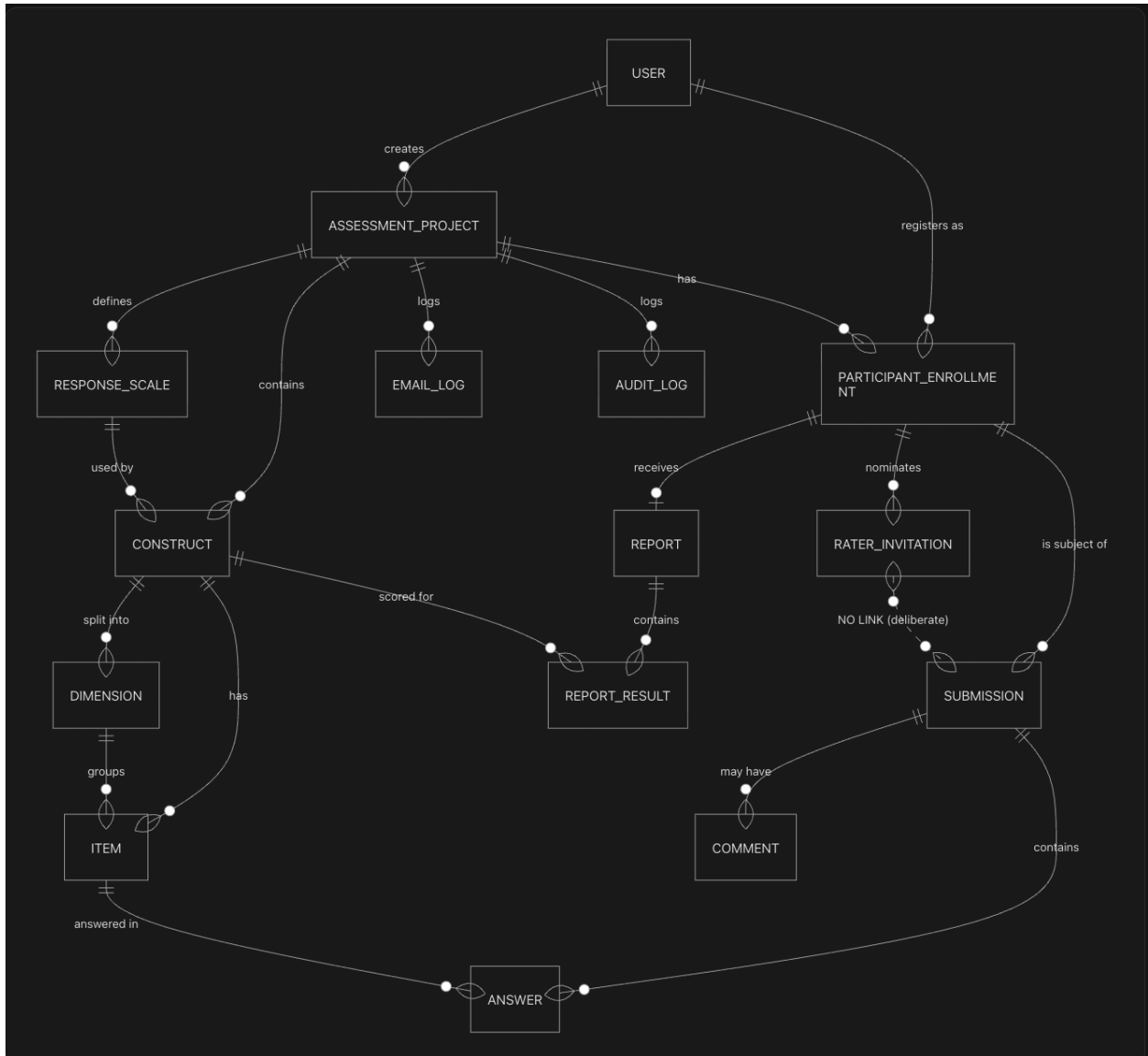
Create Survey: The screen that allows administrators to create surveys

- Accessed via the surveys tab
- Saves completed surveys to the website's database.

Survey View: The screen displayed after a user/administrator clicks on the survey they wish to view or complete.

Data / Database / Storage Design:

ER diagram:



What is stored:

Accounts	admin and participant emails, password hashes
Project and survey configuration	constructs, items, scales, scoring rules, thresholds.
Process and identity data	participants, rater emails, relationships, tokens, consent timestamps, reminder counts.
Response data	submissions and item answers.
Results	generated report values, review and release status.
Operational	email log, audit log.

Glossary

- **360-Degree Feedback:** A multi-source feedback evaluation model where participants complete a self-assessment and receive peer-based feedback from multiple raters to identify personal strengths and areas for growth.
- **Administrator:** A system role—typically professors or instructors—authorized to create assessment projects, build survey instruments, assign cohorts, and monitor evaluation progress.
- **Cohort:** A designated group of students or participants assigned together to a specific assessment project for a given course or semester.
- **Minimum-Rater Threshold:** A privacy and confidentiality rule requiring a minimum number of completed evaluations (default is 3) before aggregate feedback is displayed, ensuring individual rater responses cannot be identified.
- **Participant (Student):** The primary user evaluated within the system who completes a self-assessment, invites raters, and views aggregated feedback reports.
- **Rater (Peer):** An individual invited by a participant to evaluate the participant's leadership and business skills through third-person survey questions.
- **Standard Deviation:** A statistical calculation included in participant reports to show the variance and level of agreement among raters' scores for a given construct.
- **Survey Agnostic:** A flexible architectural design that enables administrators to create, modify, and configure dynamic survey templates, scoring rules, and constructs without requiring source code changes.

Error and Edge-Case Behaviour

The DataPulse360 system is designed to provide clear feedback to users when errors occur and to prevent data loss whenever possible.

Condition: User enters an incorrect username or password.

System Behavior:

- Authentication request is rejected.
- User remains on the Sign In screen.
- Error message displayed: "Invalid username or password. Please try again."
- Failed login attempt is recorded in the audit log.
- After multiple failed attempts, the account may be temporarily locked for security purposes.

Condition: A standard user attempts to access administrator-only functionality.

System Behavior:

- Request is denied.
- User is redirected to an authorized page.
- Event is recorded in the audit log.
 - Message displayed: "You do not have permission to access this resource."

Condition: User attempts to submit a survey without answering all required questions.

System Behavior:

- Submission is blocked.
- Unanswered questions are highlighted.
- Message displayed: "Please answer all required questions before submitting."

References:

Node.js Documentation. OpenJS Foundation. Available at: <https://nodejs.org/docs>

Chart.js Documentation. Available at: <https://www.chartjs.org/docs>

OWASP Web Application Security Guidelines. Available at: <https://owasp.org>

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